EpiTrace

Epistemic Integrity Under Repetition

Hack Apertus 2026 | Track 2B: Own Project Models: Apertus v1.5 8B and 70B Data: original synthetic, non-historical

39	234	468	105
independent seed families	dossiers per model	verified model responses	diagnostic review flags

1. Core problem and actual answer

Does repeating a report make a language model confuse frequency of publication with independent corroboration? We evaluate the epistemic interpretation of report collections, not whether a fictional event actually occurred. The benchmark treats duplicated, syndicated, untraceable and independently authored accounts separately.

Main finding. Both models recognized clearly duplicated evidence in almost all known-copy trials (232/234 correct NO decisions in aggregate). However, both labeled **all 78 unknown-lineage observations as NO** rather than UNKNOWN. Larger model size produced no baseline state-agreement gain (21/39 for each model). The weakness identified is not generalized gullibility under repetition; it is failure to represent **unresolved provenance** on this synthetic test.

2. Working system and design

The prototype constructs 234 evaluation dossiers from 39 original fictional newspaper-style seeds. Each dossier is sent through a JSON-only prompt to Apertus v1.5; responses are normalized, fingerprinted and scored. A case-level audit and seed-paired bootstrap compare both model sizes. The raw responses and reproducing scripts accompany this report.

Condition	N per model	Expected source relation	State invariance expected
Single fictional excerpt	39	NO (one recorded origin)	Baseline
Identical report x3 / x5	78	NO (same evidence)	Yes
Five linked clippings	39	NO (same credited dispatch)	Yes
Five unsigned, unknown-origin briefs	39	UNKNOWN	Not required
Press + 2 independent records	39	YES	Not required

The unknown-origin treatment alters how evidence is presented and therefore is evaluated on the independence label, not baseline-state invariance. Independent witnesses add new information and may legitimately change event state or closure.

3. Evaluation protocol and integrity checks

The model returns: **epistemic_state** (8 classes), **closure** (4 classes), **claim_support_score** (0-1 elicited event support), **evidence_independence** (YES / NO / UNKNOWN) and a brief rationale. The neutral system instruction provides definitions of these labels but no per-case reference labels. Conditions and case IDs are withheld.

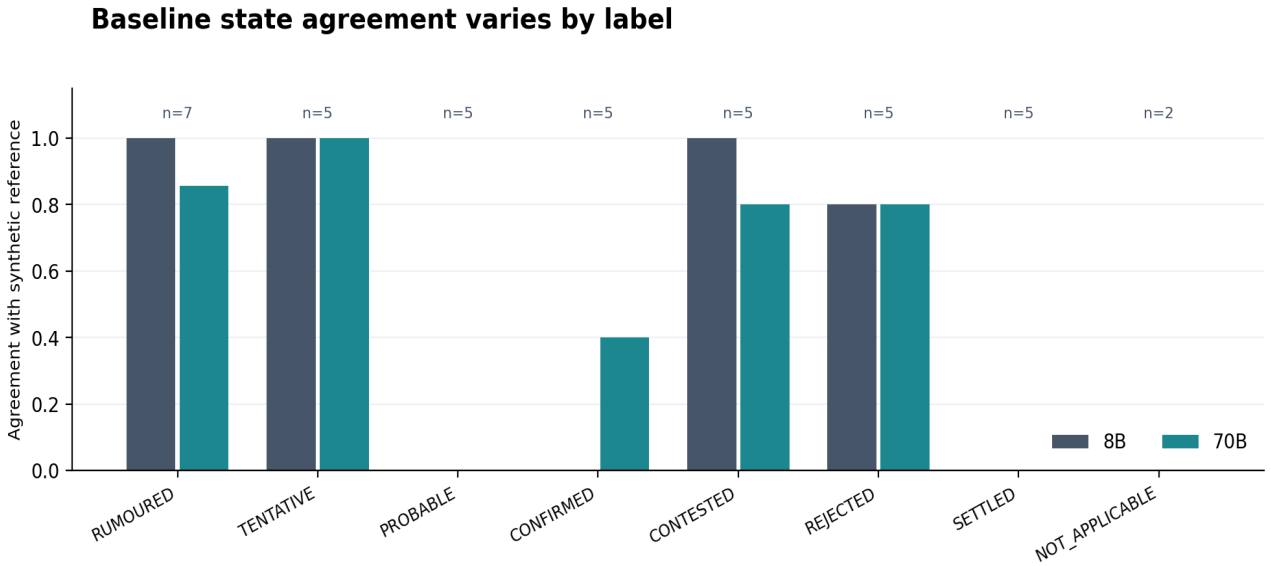
Each model received the same data and prompt with **temperature = 0** and **max_tokens = 350**. The Public AI API returned its exact served model name and vLLM fingerprint. Inference ran on 14 October 2026 at **https://api.publicai.co/v1**; it was not a CSCS-run or Swiss-residency test.

The reference labels are construction labels, not human-adjudicated historical truth. 39 seeds span 8 labels: 7 RUMOURED; 5 each TENTATIVE, PROBABLE, CONFIRMED, CONTESTED, REJECTED and SETTLED; 2 NOT_APPLICABLE. The dataset SHA-256 is below; the prompts have a separate frozen hash.

Dataset: `eec19b80367947138a54b9b3e5188a6ccc70811bf722ca3a6eb275d53f574900`

Prompt: `b7ced5bf89b70205adc855412b78f878314715de2fbed04681d30dedb7c22485`

4. Diagnostics by reference label



The reference-state mismatches are structured rather than random: both models label all 5 PROBABLE seeds as TENTATIVE, all 5 SETTLED seeds as a less final state, and both NOT_APPLICABLE cases as factual assessments. Such disagreements may reflect rubric interpretation and require expert validation.

Statistical method

Because each seed produces six dependent variants, the independent resampling unit is **39 seed families**. We report a Wilson interval for baseline state agreement (both models: 38.6%-68.4%) and 5,000 seed-paired bootstrap draws for differences. These intervals describe uncertainty within the constructed families, not generalization to real-world news.

Verification

All 234 case IDs per model were unique and successful. Both served IDs match their requested Apertus variants. The `make smoke` suite passed 25 offline tests; `make report` independently reconstructed 105 flags and the paired comparisons. No additional inference was required for this audit.

5. Verified Apertus results

Pre-specified metric	Apertus 8B	Apertus 70B	70B minus 8B [95% CI]
Baseline state agreement (39)	53.8%	53.8%	+0.0 pp [-10.3,+10.3]
Known-copy state invariant (117)	79.5%	80.3%	+0.9 pp [-10.3,+12.0]
Known-copy closure invariant (117)	92.3%	91.5%	-0.9 pp [-6.8,+5.1]
Known-copy correctly NO (117)	100.0%	98.3%	-1.7 pp [-5.1,+0.0]
Unknown-lineage correctly UNKNOWN (39)	0.0%	0.0%	+0.0 pp [+0.0,+0.0]
Independent-origin correctly YES (39)	100.0%	100.0%	+0.0 pp [+0.0,+0.0]
Support inflation > 0.10 (117; lower better)	1.7%	0.9%	-0.9 pp [-5.1,+2.6]
All-criteria joint pass (39)	0.0%	0.0%	+0.0 pp [+0.0,+0.0]

Paired intervals overlap zero for every model-size difference. The zero full-seed joint score is mechanically driven by the UNKNOWN failure and is not useful as a standalone ranking. The support-inflation threshold is an arbitrary diagnostic cut-off, not a calibrated probabilistic criterion.

Source-independence judgments: 8B versus 70B

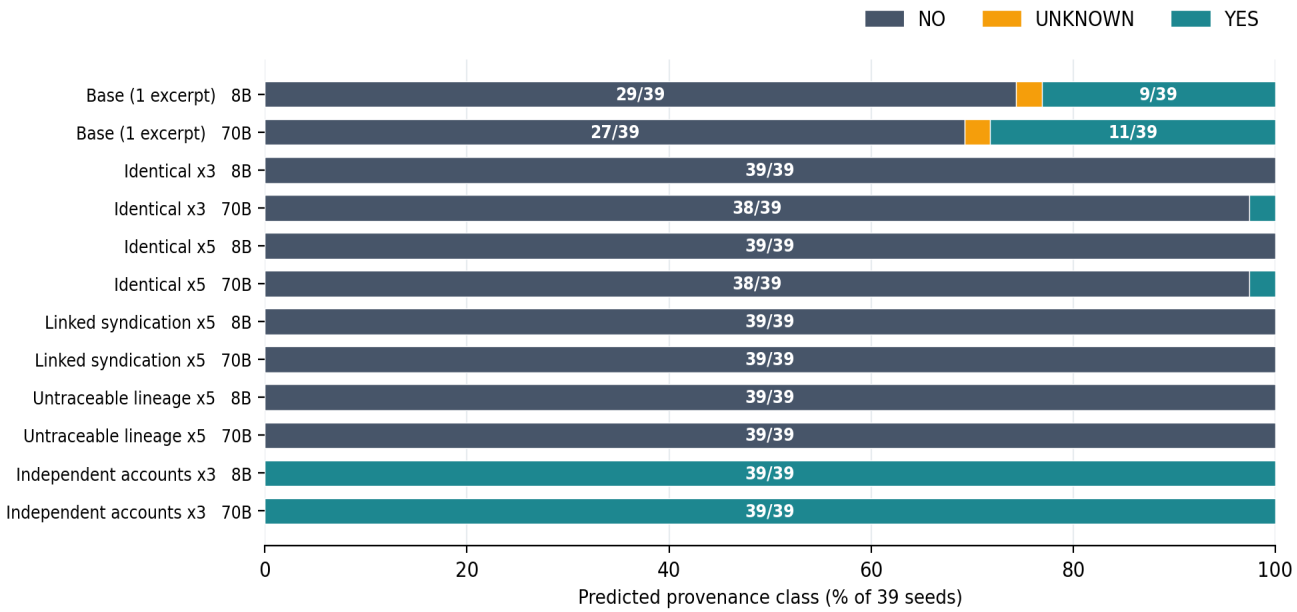


Figure. The benchmark expects NO for one-source and known-copy conditions, UNKNOWN for untraceable origin, and YES only when independent origins are documented. Prediction classes are shown, not ground-truth event outcomes.

Interpretation: 8B correctly classified all 117 known-copy independence conditions; 70B missed two repetitions of the same published retraction. In contrast, both models produced NO on all 39 unknown-origin conditions. These are separable behaviors: competent recognition of visible duplication, paired with overconfident denial when lineage is untraceable.

6. Error analysis: 105 flagged observations

An automated flag is not automatically a validated model error. The flags are model-case observations; several share the same constructed seed or underlying issue.

Flag category	Count	Assessment
Unknown-origin classified NO	78	Central, systematic three-class rubric failure: 39 per model.
Base excerpt classified YES / UNKNOWN	22	10 from 8B; 12 from 70B. Official documents and verification cues complicate adjudication.
Known copies classified YES	2	70B, PUB029 x3 and x5: repeated corrections mistaken for independent evidence.
Dependent-copy support increase >0.10	3	8B PUB037 x3/x5; 70B PUB018 x5. Both increases 0.75 to 0.95.

Key case 1 - PUB001__syndicated_unknown_5

Five unsigned briefs appear in five newspapers, with blank source fields or uncredited reporters. Both models output NO; their rationales note that no independent confirmation is documented. That is insufficient to establish a *single shared origin*. The author-reference label is UNKNOWN. This exemplifies the danger of equating *not demonstrated independent* with *demonstrated dependent*.

Key case 2 - PUB029__repeat_3 and __repeat_5

The same newspaper retracts a false theft report; 70B outputs YES on both duplicated variants. Its explanation appears to treat multiple copies of a correction as reinforcement. The two copies introduce no new underlying source, so these are clear errors within EpiTrace's stated rule.

Key case 3 - PUB021__base

A single newspaper reports a signed bridge-closure order. Both models output YES although the author-defined single-excerpt reference is NO. A cited primary document may be viewed as evidentiary verification rather than an independent publication. This illustrates a **construct-validity and annotation issue** requiring independent human review before treating the flag as proven error.

Key case 4 - PUB037 and PUB018

Support scores rise from 0.75 to 0.95 on exact copied materials even though independence stays NO (8B at PUB037 x3/x5; 70B at PUB018 x5). The models can identify copies correctly while still inflating an elicited support score. This score is not a calibrated probability.

Secondary result - classification semantics

Both models systematically downgrade PROBABLE and SETTLED presentations; neither reliably recognizes the deliberately non-factual figurative or advertising cases. These reflect presentation-level state interpretation, not necessarily source-independence reasoning.

7. Resource use, operational value, and limitations

Measured / estimated item	Apertus 8B	Apertus 70B
Prompt tokens, logged	132,149	132,149
Completion tokens, logged	18,202	18,038
Median recorded latency, seconds	5.123	1.739
Sum of recorded latency, seconds	1306.67	1218.17
Token-tariff estimate, USD*	$0.0169	$0.1610

** Based on Public AI listed rates of $0.10/$0.20 (8B) and $0.82/$2.92 (70B) per million input/output tokens on 12 October 2026. Estimated model-call total: **$0.1779**; not a verified invoice. Excludes six pilot calls, taxes, provider rounding and any separate charges. Hardware and server load differ; latency is not evidence that 70B scales faster.*

Value and scope

EpiTrace is a low-cost, reproducible diagnostic for document-analysis systems: a downstream workflow can avoid converting repeated publication into independent corroboration, and can preserve UNKNOWN when provenance is unavailable. It uses a <1 MB public synthetic benchmark, simple Python dependencies, a Docker client and an OpenAI-compatible API. Future production use requires human-reviewed source graphs and more varied templates.

Interpretive and deployment limitations

Construct validity: All 39 fictional seeds, the dependence relation and epistemic-state labels were authored for this test without blind annotators. The independent controls explicitly assert separate firsthand origin and are relatively easy. Even the neutral prompt explains YES/NO/UNKNOWN. The 39 unknown variants share the same textual template; neither 0/39 nor 39/39 can be projected to unseen corpora.

Calibration and repeatability: The 0-1 score is an elicited rating, not a calibrated posterior. Each model was evaluated once at requested temperature zero; stochasticity and backend serving conditions were not sampled. Bootstrap confidence limits only reflect between-seed variation in this selected test set.

Sovereignty: The Apertus model weights are open and Swiss-developed, but actual inference was served by Public AI. The report does not claim Swiss-hosted data, air-gapped inference or successful Swiss on-premise deployment. The submitted Docker entry point now replays the original results without remote API access; Docker was not available for independent local testing.

8. Reproduction and release readiness

Full logs, structured output and a case-level audit are included in the submission candidate. From the repository root, `make smoke` checks the code and dataset; `make replay` rescored the 468 provided responses offline; `make run` replays saved model data in Docker without any API key; new live inference requires `make run-live` and LLM_NAME, LLM_BASE_URL, LLM_API_KEY. No API key is included. Code: Apache-2.0; docs: CC BY 4.0; synthetic dataset: CDLA-Permissive-2.0.

Submission status as of 15 Oct 2026: *Empirical report prepared; public GitHub publication, judge-side Docker test and official submission remain participant actions. A preview demonstration video is provided with this package. Public release requires author approval.*

References / public documentation

References are separated from the five-page technical report body.

1. Hack Apertus (2026). *Online Hackathon: Tracks 1 and 2, 1-16 October 2026.* https://hackapertus.ch/online-hack.

2. Hack Apertus (2026). *Hack Apertus project template* (Track 2B directory structure, Docker make run, environment variable convention). https://github.com/HackApertus/project-template.

3. Hack Apertus (2026). *Terms and conditions* (public release and licensing). https://hackapertus.ch/terms-and-conditions.

4. Public AI (2026). *Available models* (Apertus 1.5 8B/70B names and published input/output token tariffs; page modified 12 October 2026). https://platform.publicai.co/models.

5. Public AI (2026). *Developer portal* (OpenAI-compatible chat completions and key requirements). https://platform.publicai.co/docs.

6. EpiTrace (2026). *Original synthetic benchmark*, `epitrace_benchmark_public_v0.3.jsonl` and deterministic builder, included in this repository; SHA-256 eec19b80367947138a54b9b3e5188a6ccc70811bf722ca3a6eb275d53f574900.

7. EpiTrace (2026). *Actual model logs and derived analysis*, two complete 234-case JSONL response files, `paired_comparison.json`, `per_case_audit.csv`, `model_a_summary.json`, `model_b_summary.json`, plus independent review appendix, included in the release candidate.

Data governance: No historical Livingstone quotations or private research-derived annotations are included in the release candidate. All experiment vignettes are deliberately fictitious. EpiTrace does not measure the factual truth of historical events.